Linux `find` Command Notes

The `find` command is used to **search for files and directories** in a directory hierarchy based on different conditions like name, type, size, permissions, and more.

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1. Basic Search by Name

```
find /path -name "file.txt"
```

Description: Searches for `file.txt` inside `/path` and its subdirectories.

Example:

```
find /home -name "notes.txt"
```

Output:

```
/home/prashant/Documents/notes.txt
```

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2. Case-Insensitive Search

```
find /path -iname "file.txt"
```

Description: Finds files ignoring case (e.g., matches `File.txt`, `FILE.TXT`).

Example:

```
find /home -iname "notes.txt"
```

Output:

```
/home/prashant/notes.txt
/home/prashant/Notes.TXT
```

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3. Search by File Type

```
find /path -type f   # Search for files
find /path -type d   # Search for directories
```

Example:

```
find /etc -type d -name "nginx"
```

Output:

```
/etc/nginx
```

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4. Search by Size

```
find /path -size +10M   # Files larger than 10MB
find /path -size -100k  # Files smaller than 100KB
```

Example:

```
find /var/log -size +50M
```

Output:

```
/var/log/syslog.1
```

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5. Search by Modified Time

```
find /path -mtime -7   # Modified in the last 7 days
find /path -mtime +30  # Modified more than 30 days ago
```

Example:

```
find /home -mtime -1
```

Output:

```
/home/prashant/report.txt
```

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6. Search by Permissions

```
find /path -perm 644   # Files with exact 644 permissions
find /path -perm /111  # Files executable by someone
```

Example:

```
find /usr/bin -perm /111
```

Output:

```
/usr/bin/ls
/usr/bin/cp
```

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7. Execute Command on Found Files

```
find /path -name "*.log" -exec rm {} \;
```

Description: Deletes all `.log` files under `/path`.

Example:

```
find /tmp -name "*.tmp" -exec rm {} \;
```

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8. Find Empty Files or Directories

```
find /path -type f -empty   # Empty files
find /path -type d -empty   # Empty directories
```

Example:

```
find /home/prashant -type f -empty
```

Output:

```
/home/prashant/empty.txt
```

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9. Combine Multiple Conditions

```
find /path -type f -name "*.txt" -size +1M
```

Description: Finds `.txt` files larger than 1MB.

Example:

```
find /home -type f -name "*.log" -mtime -2
```

Output:

```
/home/prashant/app.log
```

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10. Limit Search Depth

```
find /path -maxdepth 1 -name "*.txt"
```

Description: Searches only in the current directory, not subdirectories.

Example:

```
find /home/prashant -maxdepth 1 -name "*.txt"
```

Output:

```
/home/prashant/todo.txt
```

Quick Summary of Common Options

Option	Meaning
`-name`	Search by file name (case-sensitive)
`-iname`	Case-insensitive search
`-type`	File type (f=file, d=dir, l=link, etc.)
`-size`	File size (+ = greater, - = smaller)
`-mtime`	File modified time

Option	Meaning
`-perm`	File permissions
`-exec`	Run a command on each match
`-empty`	Find empty files or directories
`-maxdepth`	Limit directory depth

✅The `find` command is a **powerful search tool** in Linux for locating files and automating tasks on them.